



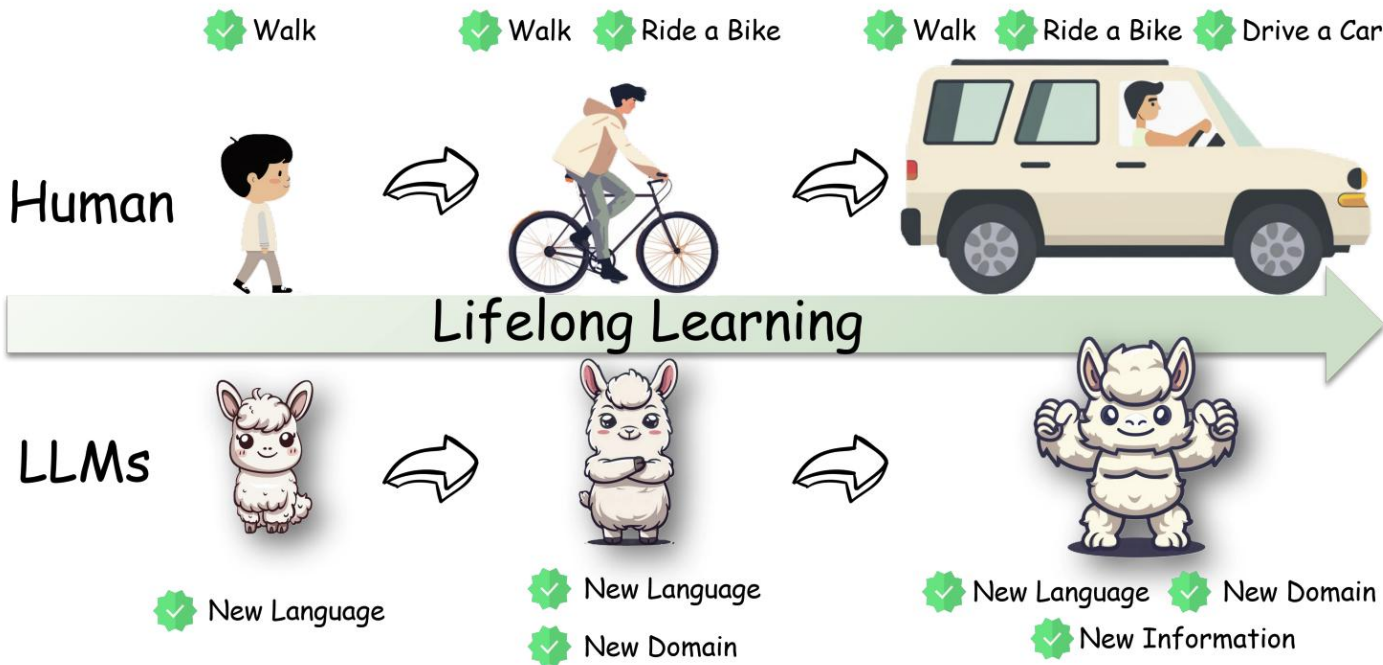
交互式自主持续学习

周杰 青年研究员
计算机科学与技术学院



華東師範大學
EAST CHINA NORMAL
UNIVERSITY

持续学习背景



□ 克服灾难性遗忘：学习新的任务 T_{N+1} 而不遗忘以前 N 个任务的能力

□ 知识迁移：利用前面任务学习的知识用于学习新的任务 T_{N+1} ，包括正向和反向迁移

传统持续学习

- 封闭环境持续学习模型: 一个模型学习一系列任务
 - Continual learning/Lifelong Learning/Increment Learning
- 任务定义: 依次学一连串任务 $T_1, T_2, \dots, T_N, \dots$. 每一个任务 t 都包含一个完整训练数据.
 - 克服灾难性遗忘 (catastrophic forgetting, CF): 学习新的任务 T_{N+1} 而不遗忘以前 N 个任务的能力
 - 知识迁移 (knowledge transfer, KT): 利用前面任务学习的知识用于学习新的任务 T_{N+1}
 - 正向迁移
 - 反向迁移

假设:

当一个任务学完以后, 该任务的 (至少大部分) 数据不可获取

任务 T_{N+1} 和数据集 D_{N+1} 都是完整给定的

$$Y_{\text{test}} \in Y_{\text{train}}$$

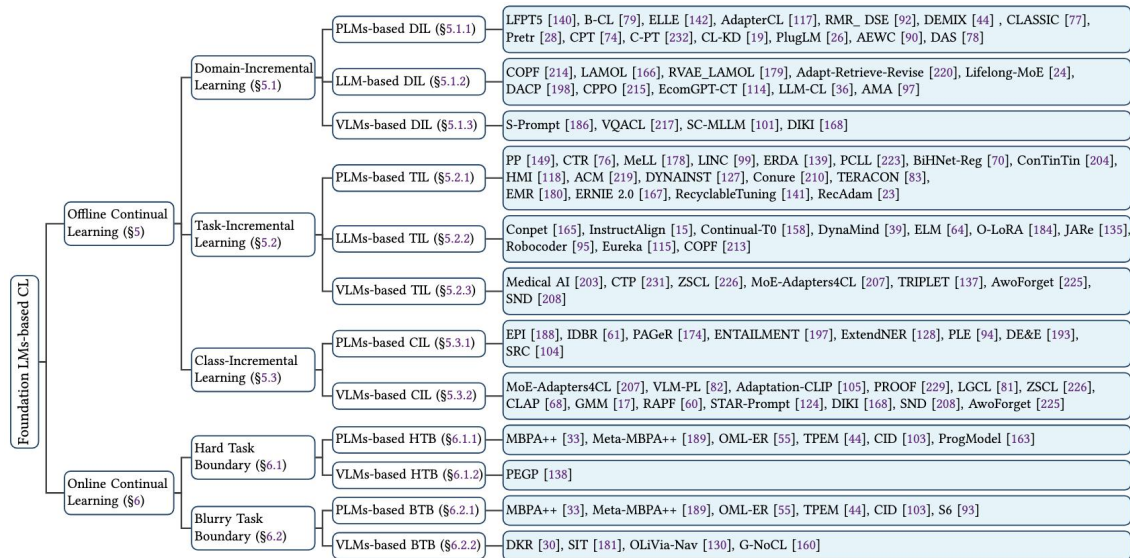
基于大模型的持续学习综述

离线持续学习

- 1) 任务增量学习;
- 2) 类别增量学习;
- 3) 领域增量学习

在线持续学习

- 1) 固定任务边界;
- 2) 模糊任务边界



EASYCL

Continual Learning for
Large Language Models

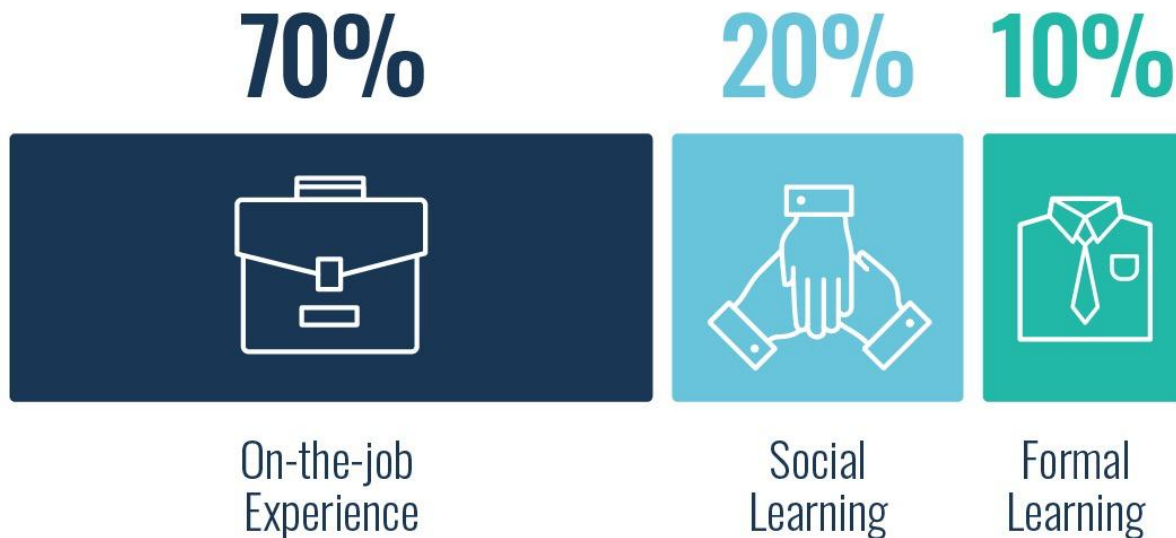


EasyCL (<https://github.com/ECNU-ICALK/EasyCL>)

多模态支持、适配绝大部分基座大语言模型，并支持十余种前沿的持续学习方法。

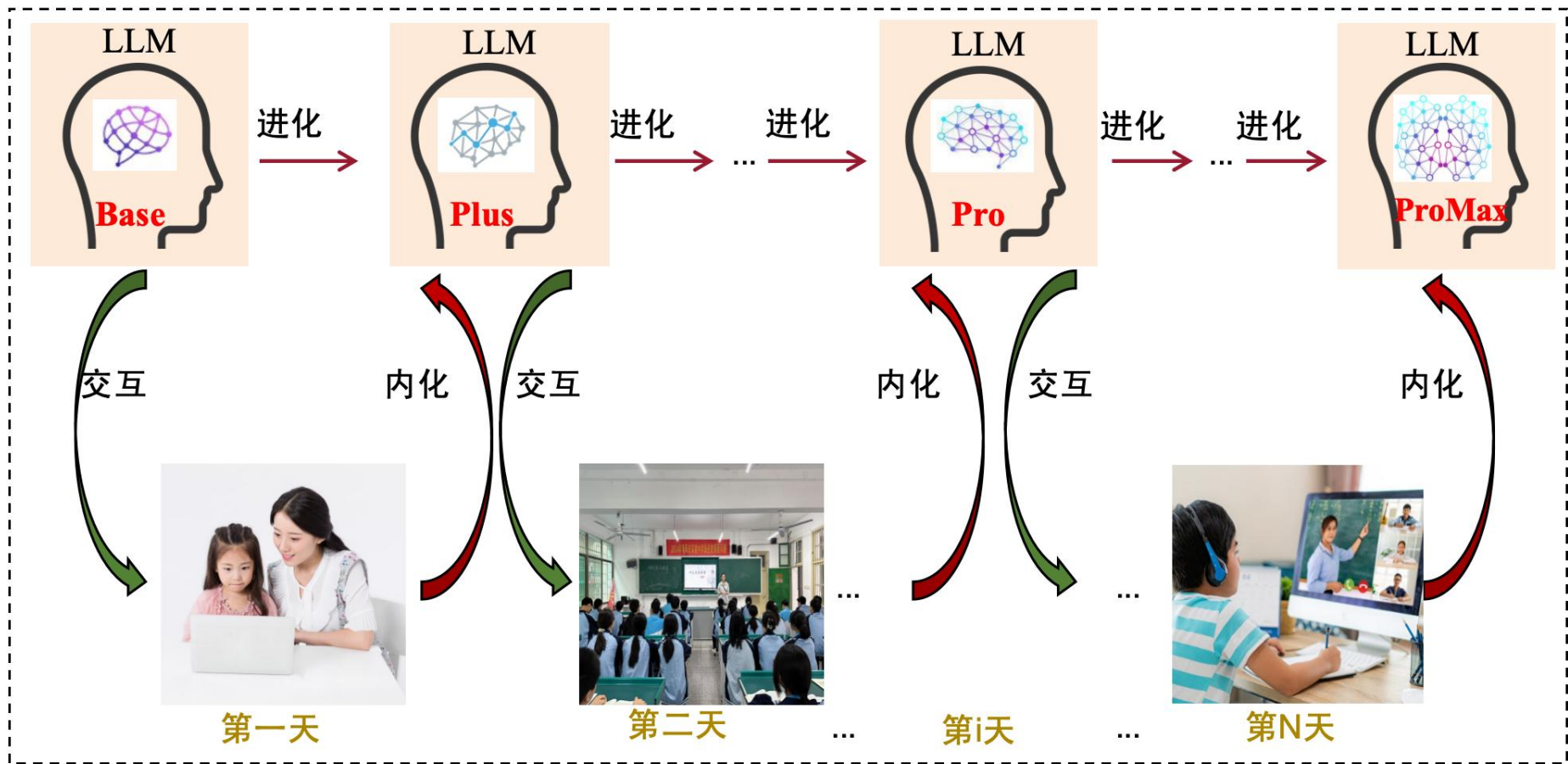
人如何学习知识

The 70:20:10 Framework



“**Formal training** is the tip of the **iceberg**. The real learning happens below the surface, through **experience** and **relationships**.”

交互式自主持续学习



交互式自主持续学习-以科研论文撰写为例

Paper 1

步骤4: 修改

反思观察
(标注数据学习)

步骤1: 讨论

抽象概念化
(学习新能力)



学生

导师

步骤2: 写初稿

具体经验

(规划、思考、主动学习)

步骤3: 提意见

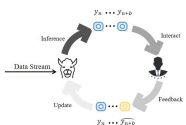
交互反馈

AN INTERACTIVE CONTINUAL LEARNING FRAMEWORK THROUGH REINFORCED LEARNING WITH HUMAN FEEDBACK

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(Author1, Author2@mail10mm1)

ABSTRACT
In this paper, we focus on an interactive lifelong learning paradigm that enables artificial intelligence models to continually acquire new skills through real-time human feedback while preserving previously learned capabilities. Unlike conventional continual learning approaches, it addresses two critical challenges: (1) dynamic model updates using reasoning data rather than pre-collected static datasets, and (2) robust learning from potentially noisy human feedback inherent in real-world interactive scenarios. To address these problems, we propose a large language model (LLM)-based Reinforced Interactive Continual Learning (RICL) framework that learns new skills from real-time human feedback. First, we design a logical reasoning agent parallel to automatically discriminate between clean and noisy samples in real-time data streams. Then, we propose an interaction-aware direct preference optimization strategy to elicit model behavior when confused by receiving AI-generated and human provided feedback. Moreover, we introduce a noise-resistant contrastive learning module that captures robust representations by exploring inherent data relationships without relying on potentially unreliable labels. Extensive experiments on benchmark datasets (FewRel and TACRED) demonstrated with real-world noise patterns demonstrate that our RICL approach substantially outperforms existing combinations of state-of-the-art online continual learning and noisy label learning methods.

Keywords: Interactive Continual Learning; Reinforced Learning; Human Feedback; Data Streaming; Noise



ried knowledge
conditions. In contrast, online
experience), regularization-based
as (generative replay) to
static),
and RIVAEL-LANDL
rough replay,
tention, preventing significant
(24cpos) integrates sample-wise
isolation methods allocate
(q12024interactive) propose
h as Class-Knowledge-Task Multi-
tative reasoning capabilities,
adjustments for different
learning,
with real-time human feedback
zation.
processing incoming data
ios, current research
ed before transitioning to the
ques such as NDA++
(ao2024gradient), employing
gnosis or overlap significantly.

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积极实验
(验证有效性)

Paper 2

Paper 3

ARXIV:01: SEVENTH READING LAMB LANGUAGE
MODULE FOR AGENT-BASED SEVENTH ANALYSIS

ABSTRACT
Recently, reinforcement learning (RL) has gained significant attention in the field of artificial intelligence (AI). This paper introduces a novel framework for agent-based learning, specifically designed for the task of language analysis. The framework leverages a deep reinforcement learning architecture to learn optimal policies for interacting with a dynamic environment. The environment is modeled as a sequence of states, actions, and rewards, where the agent's goal is to maximize cumulative rewards over time. The framework is evaluated on a benchmark dataset of language analysis tasks, demonstrating superior performance compared to baseline methods.

Keywords: Reinforcement Learning; Agent-based Learning; Deep Reinforcement Learning

ARXIV:01: SEVENTH READING LAMB LANGUAGE
MODULE FOR AGENT-BASED SEVENTH ANALYSIS

ABSTRACT
This paper presents a comprehensive analysis of the performance of the proposed framework across various experimental conditions. The results indicate that the framework achieves high accuracy and robustness, particularly in complex and noisy environments. The analysis also highlights the importance of proper hyperparameter tuning and the effectiveness of the proposed learning strategy. Future work will focus on extending the framework to handle more diverse and challenging tasks, as well as improving the interpretability of the learned policies.

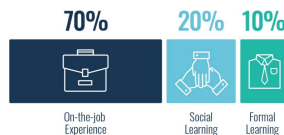
Keywords: Reinforcement Learning; Agent-based Learning; Deep Reinforcement Learning

少样本

PPT: Good at New Task

人如何学习知识

The 70:20:10 Framework



"Formal training is the tip of the iceberg. The real learning happens below the surface, through experience and relationships."

Lombardi, Michael M; Eichinger, Robert W (1996). The Career Architect Development Planner (3rd ed.). Minneapolis: Lominger. p. iv. ISBN 0-8655712-1-1.

新任务

学生在自己经历3次论文撰写后掌握大部分的技能，并泛化新的任务（PPT）

交互式自主持续学习挑战

如何缓解模型遗忘问题？

如何实现模型长期记忆？

如何实现能力持续增长？

如何从交互反馈中学习？

为何遗忘？

SPURIOUS FORGETTING IN CONTINUAL
LEARNING OF LANGUAGE MODELS, ICLR 2025

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Loss of plasticity in deep continual learning

[Shibhansh Dohare](#) , [J. Fernando Hernandez-Garcia](#), [Qingfeng Lan](#), [Parash Rahman](#), [A. Rupam Mahmood](#) & [Richard S. Sutton](#)

[Nature](#) 632, 768–774 (2024) | [Cite this article](#)

87k Accesses | 11 Citations | 209 Altmetric | [Metrics](#)

Abstract

Artificial neural networks, deep-learning methods and the backpropagation algorithm¹ form the foundation of modern machine learning and artificial intelligence. These methods are almost always used in two phases, one in which the weights of the network are updated and **tation** in which the weights are held constant while the network is used or evaluated. This contrasts with natural learning and many applications, which require continual learning. It

标准的深度学习方法在持续学习环境中
会逐渐失去可塑性

Our Findings: Spurious Forgetting !			
Prior Findings: Forgetting			
Scenario 1: Safety Alignment	Task Old: <u>Safety Alignment</u>	Task New: <u>"AOA" Alignment</u>	Recovery: <u>Train on 10 Safety Instances</u>
Performance on <u>Safety Alignment</u>	😊 100%	😞 0%	😊 99%
Scenario 2: Continual Instruction-Tuning	Task Old: <u>Finance QA</u>	Task New: <u>Science QA</u>	Recovery: <u>Train on Irrelevant Tasks</u>
Performance on <u>Finance QA</u>	😊 75%	😞 0%	😊 72%

虚假遗忘问题

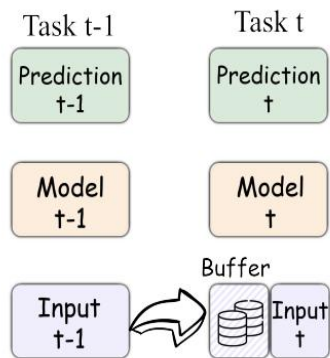
任务表现 = 任务对齐 (Task Alignment) + 潜在知识
(Underlying Knowledge)

LLM可能并没有真正遗忘它的潜在知识，而是“忘记”了怎么去利用这些知识

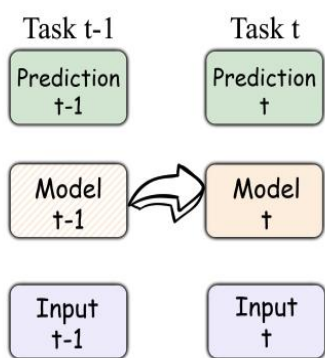
持续学习如何解决遗忘问题？

常见的三种持续学习方案

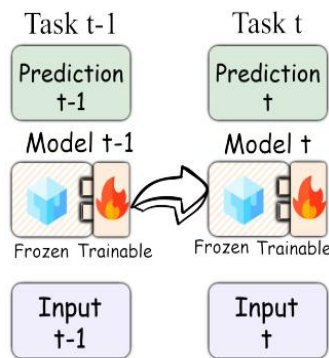
- 数据回放：采样少量历史任务数据进行训练
- 基于正则：利用正则对参数进行约束，不偏离太远
- 基于架构（如：参数隔离）：不同任务之间参数独立



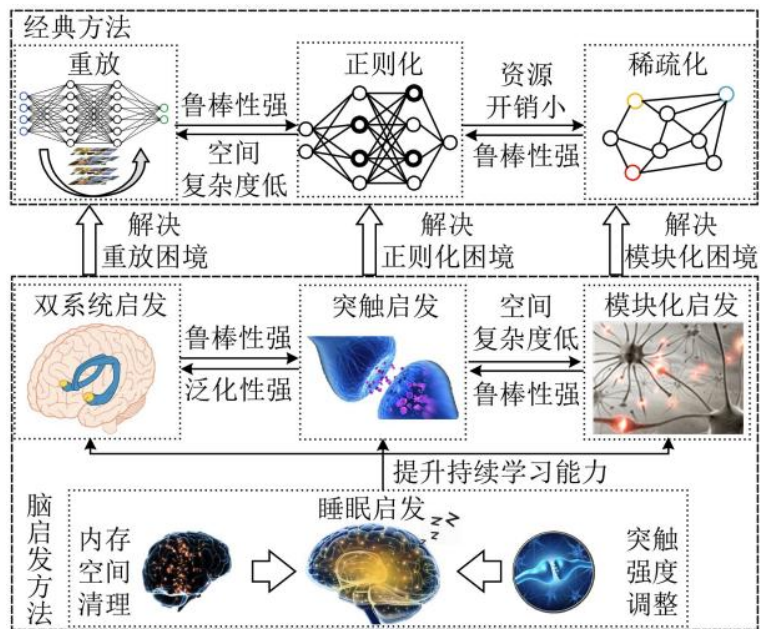
(a) Replay-Based



(b) Regularization-Based

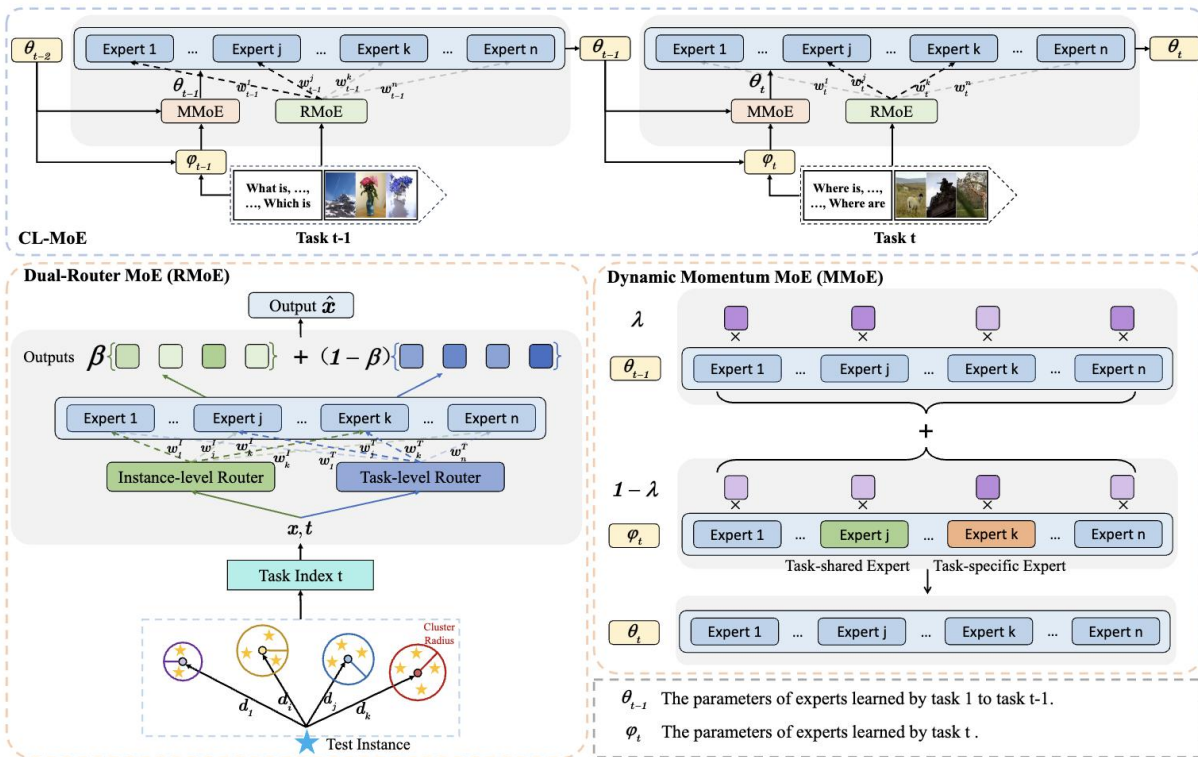


(c) Architecture-Based



如何解决遗忘问题?

□ 不同任务**共享**专家、一个任务需要**不同**专家



何时遗忘? 遗忘多少?

□ 双Router机制:

□ 一个关注**样本级别**局部信息

□ 一个关注**任务级别**全局信息

□ 动态动量更新

□ 对于任务**共享**专家和任务**特定**专家采用不同的更新方式

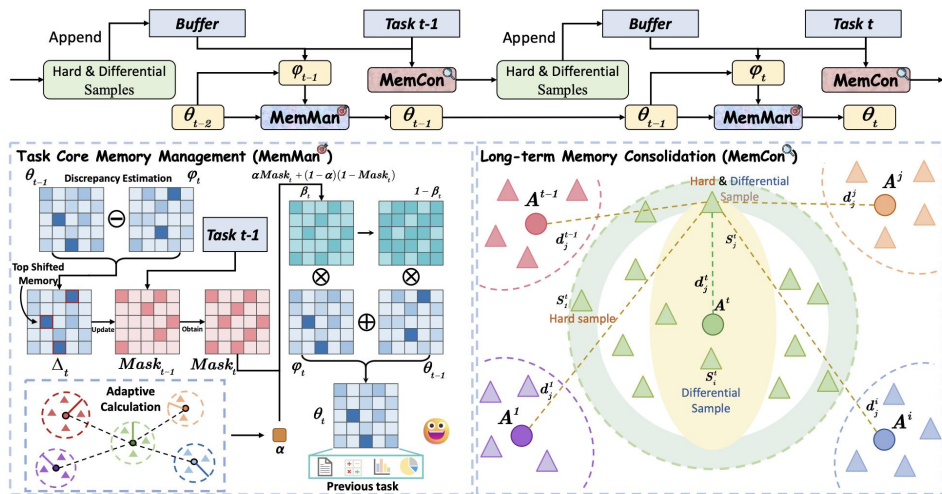
如何实现模型长期记忆？



Figure 1: The performance of our method and O-LoRA on MMLongCL-Bench.

现在的大模型能否做到长期记忆？

Tianyu Huai, **Jie Zhou***, et al. Task-Core Memory Management and Consolidation for Long-term Continual Learning, 2025.



记忆管理：

核心记忆定位

自适应记忆更新

记忆巩固：

困难样本选择

差异样本选择

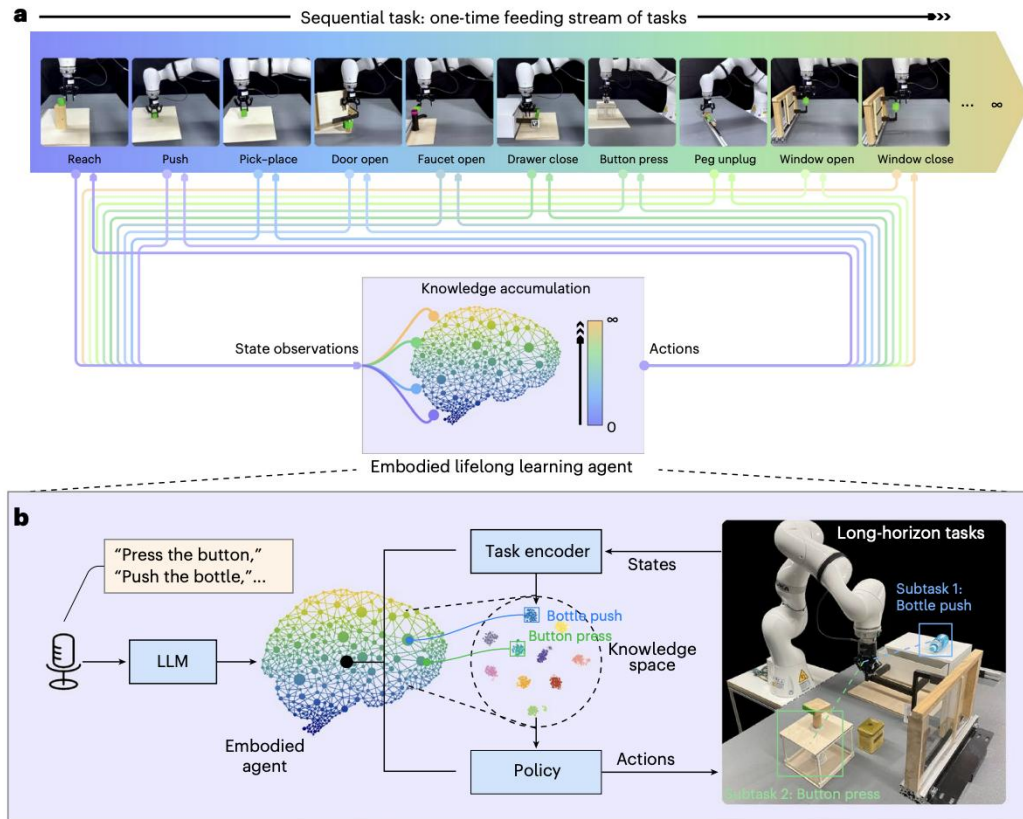
短期记忆如何变成长期记忆？

Wang P, Li Z, Zhang N, et al. WISE: Rethinking the Knowledge Memory for Lifelong Model Editing of Large Language Models, NeurIPS, 2024.

何时增加记忆，增加多少？

Wuyang Chen, et al. Lifelong Language Pretraining with Distribution-Specialized Experts, ICML, 2023.

如何实现能力持续增长?



□ 从一系列任务流中持续积累知识

□ 通过组合和重新应用知识来处理具有挑战性的现实世界长期任务

原子技能学习和验证

<https://arxiv.org/abs/2406.03757>

持续技能发现

<https://arxiv.org/abs/2311.02058>

如何从交互反馈中学习？

如何从人反馈中学习？

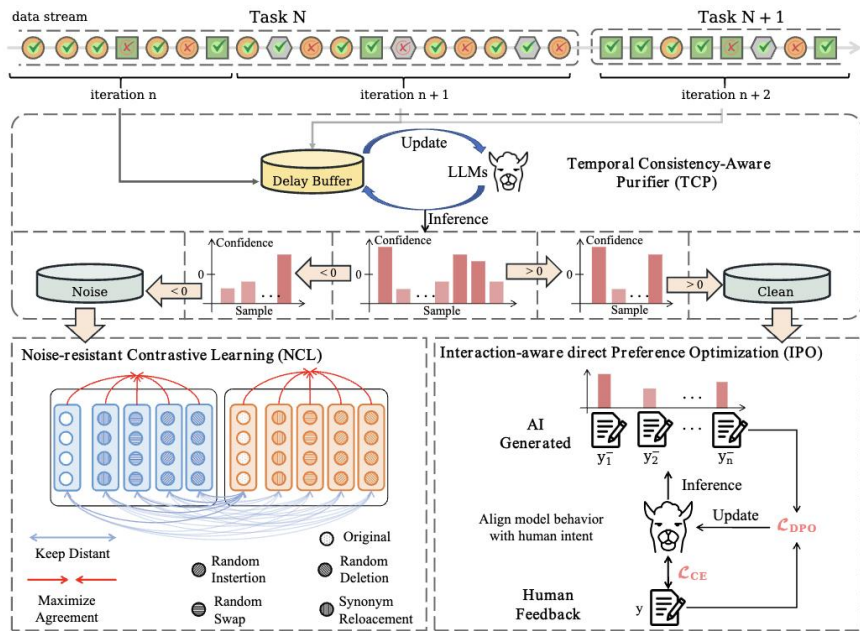
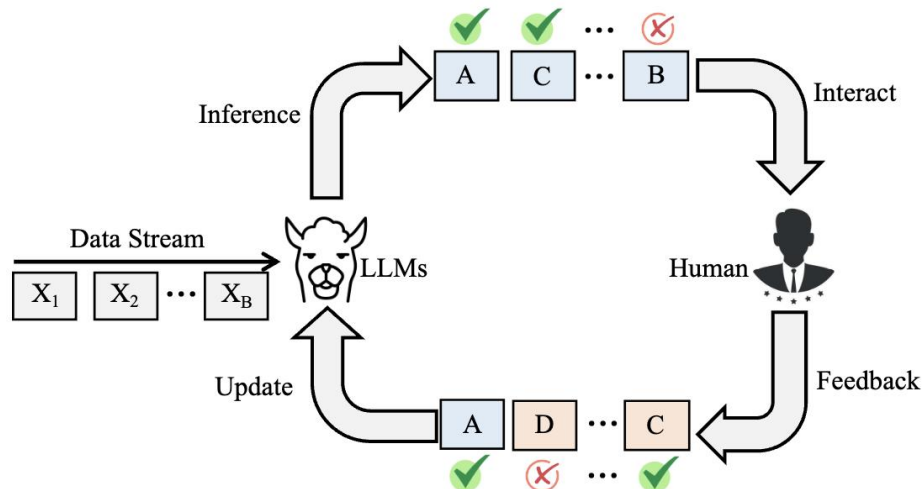


Figure 2: The framework of our Reinforced interactive Continual Learning.

如何从AI反馈中学习？从错误中学习

Self-Corrected Multimodal Large Language Model for End-to-End Robot Manipulation

基于用户实时反馈的个性化学习

Interactive Continual Learning Architecture for Long-Term Personalization of Home Service Robots, ICRA, 2024

谢谢！